

HARSHAL KHARABE

Python Developer | ETL Developer | 2+ Years of Experience

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SUMMARY :

Backend developer with experience building event-driven orchestration systems, ETL pipelines, and scalable APIs using FastAPI and PostgreSQL. Strong understanding of async workflows, system design, and enterprise data migration.

EXPERIENCE :

Python Developer Intern

Datamatter Technologies

April 2024 - April 2025

- 1.RESTful APIs for seamless data flow between backend services and the frontend.
- 2.Ensured code quality through unit testing, code reviews, and adherence to best practices for clean and maintainable code.
- 3.Designed and optimized PostgreSQL database schemas and queries to enhance performance and reliability.
- 4.Technologies Used: Python, FastAPI, PostgreSQL, Microservices etc

Backend Developer

Datamatter Technologies

May 2025 - Present

- 1.Built secure REST APIs, optimized database queries and async execution for high-throughput data processing, and implemented authentication and role-based access control.
- 2.Built backend services to **extract data from legacy systems**, apply validation and transformation rules, and load structured data into target platforms (e.g., Salesforce).
- 3.Implemented retry mechanisms, timeouts, and backoff strategies to handle partial failures, network issues, and large data volumes during migration.
- 4.Added data validation checks, schema enforcement, and audit logs to ensure **data consistency, traceability, and compliance** throughout the migration lifecycle.
- 5.Technologies Used: Python, FastAPI, Pandas, NumPy, PostgreSQL, REST APIs, AsyncIO, APScheduler, JWT, ETL Pipelines.

PROJECTS:

Event-Driven Orchestrator :

- Designed and implemented a scalable event-driven orchestrator using FastAPI and LangGraph, managing complex multi-step workflows with state persistence across failures, supporting 50+ agent types and 1,000+ concurrent executions.
- Built a robust async agent coordination system using AWS SQS, enabling non-blocking execution of long-running tasks (hours-long) via callback mechanisms, reducing system downtime by 80%.
- Engineered smart retry logic with exponential backoff and circuit-breaker patterns, achieving 95% automatic recovery from transient failures without manual intervention.
- Implemented an advanced “brake mechanism” for human-in-the-loop workflows, allowing executions to pause for manual intervention and resume seamlessly, ensuring 99.9% workflow completion rates.
- Created an agent abstraction layer decoupling orchestration from business logic, enabling independent scaling and deployment of 20+ microservices with minimal cross-service dependencies.

TECHNICAL SKILLS :

Programming Languages : Python, Java, Javascript, C++

Frameworks & Libraries : FastAPI, Django, NumPy, Pandas, Matplotlib, SciPy, Scikit-learn

Database Management : MySQL, PostgreSQL, MongoDB, Supabase, Firebase

Tools & Technologies : Postman, Git-GitLab, VS Code, Jira

EDUCATION :

Bachelor of Computer Application (BCA) | Amravati University | June 2022 - June 2025

CGPA : 8.12